Collide, Orbit, or Escape: A Computational Study of Attraction Laws from Pursuit to Gravity

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1 Problem Statement

1.1 Motivation and Context

Binary star systems are among the most common structures in the universe, yet their long-term behaviour can be surprisingly diverse. Depending on the masses, separation, and initial velocities, the two bodies may settle into a bounded orbit, escape from one another, or collide. In this project, we aim to rigorously study these three outcomes through the lens of relative motion. We can write the positions of two bodies as $\mathbf{r}_1(t)$, $\mathbf{r}_2(t) \in \mathbb{R}^2$, work throughout with the relative vector

$$\mathbf{r}(t) = \mathbf{r}_1(t) - \mathbf{r}_2(t), \qquad \mathbf{r} = (x, y), \qquad \rho = \|\mathbf{r}\| = \sqrt{x^2 + y^2},$$

and ask how the structure of the attraction law pertaining to $\mathbf{r}(t)$ determines the final outcome of the binary star (two-body) system.

More specifically, we aim to contrast two idealized and standard interaction laws. In pure pursuit, one body's velocity always points toward the other. In Newtonian gravity, one body's acceleration always points toward the other. On the surface level, these two laws appear similar, as both describes "attraction toward the partner", but they produce fundamentally different motion. Understanding the structural differences of these two law, and their generalization, mathematically is thus the central motivation of the project.

1.2 Research Question

> *What structural differences between pursuit and gravitational attraction determine whether two bodies collide, orbit, or escape?*

The insight we hope to develop throughout our computational simulation is the decisive difference between first-order (velocity-level) versus second-order (force-level) attraction.

Other sub-questions we aim to explore are:

- Why does velocity-level pursuit force finite-time capture?
- Why can force-level gravity force a stable, bounded orbit instead of collapsing?
- For a generalized equation of central force (force where attraction points inwards), when do we obtain collision, bounded motion, or escape?
- How do numerical simulations reveal and confirm these differences?

1.3 Why it is worth studying

The project connects a clean mathematical dichotomy grounded in ODEs to a concrete astrophysical system, whilst our aim of generalizing this observation to other central-force families poses a genuine research question with a definite answer. The scope is manageable both mathematically and computationally, whilst also serving as an extension to content not covered in the MAT244 syllabus.

2 Related Work

2.1 STARFORGE Program

One related project is STARFORGE, which stands for *STAR FORmation in Gaseous Environments*. STARFORGE is a computational simulation program designed to study star formation in gaseous environments with detailed feedback physics. This program can be used to investigate a variety of astrophysical questions, such as:

- Why do stars have the masses that they do?
- Why do stars form in clusters, and how do these clusters assemble?
- Why is star formation so slow and inefficient in many observed environments?
- How do winds, radiation, jets, and supernovae affect the gas surrounding newborn stars?
- Where do binary stars come from?

This program is relevant to our project because it provides background on the formation of stars and binary star systems. Although STARFORGE focuses mainly on star formation rather strictly on their orbital motion, it helps explain how the initial conditions of binary star systems may arise, and ground our project in real-world astrophysics applications.

2.2 J. R. Taylor, *Classical Mechanics* (University Science Books, 2005)

Provides the theoretical backbone for the gravitational and generalized models: reduction of the two-body problem to a one-body problem in the relative coordinate, conservation of energy and angular momentum, the effective potential and centrifugal barrier, and a statement of Bertrand's theorem on closed orbits. We use these results to derive our analytical predictions and mathematical models before testing them numerically.

2.3 Chases and Escapes: The Mathematics of Pursuit and Evasion (Princeton University Press, 2007)

A rigorous explanation of pursuit curves, including capture-time analysis and the moving-target case. It supports Model A and supplies the classical pursuit problems we will use as a non-trivial contrast to gravitational motion.

2.4 P. Virtanen *et al.*, "SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python," *Nature Methods* 17:261–272 (2020); and H. Rein and S.-F. Liu, "REBOUND: An open-source multi-purpose N-body code," *Astronomy & Astrophysics* 537:A128 (2012).

SciPy's `solve_ivp` provides validated ODE integrators (RK45, DOP853, and others) that we will use both directly and as a reference against our own implementations. REBOUND is an open-source project that provides high-accuracy symplectic integrators for multi-body problems we may use for cross-checking and comparison versus Euler and RK4 solvers.

- Project website: https://rebound.hanno-rein.de/

3 Proposed Work

3.1 Mathematical Models

We treat every model as a rule for the relative vector $r(t)$ and classify laws by the order at which they constrain the motion. Let $\hat{\mathbf{r}} = \mathbf{r}/\rho$.

- **Model 1: velocity-level pursuit.**

  The pursuer's velocity points directly toward a target at the origin:

$$\dot{\mathbf{r}} = -v\,\hat{\mathbf{r}}, \qquad \text{i.e.} \qquad \dot{x} = -v\,\frac{x}{\sqrt{x^2+y^2}}, \quad \dot{y} = -v\,\frac{y}{\sqrt{x^2+y^2}}. \tag{1}$$

  Let $\rho(t) = \|\mathbf{r}(t)\|$. Then

$$\dot{\rho} = \hat{\mathbf{r}}\cdot\dot{\mathbf{r}} = \hat{\mathbf{r}}\cdot(-v\,\hat{\mathbf{r}}) = -v, \qquad \Longrightarrow \qquad \rho(t) = \rho(0) - vt,$$

so capture occurs in finite time at $T = \rho(0)/v$. Because the velocity is forced to be radial, the tangential velocity, and hence the angular momentum, is identically zero.

Therefore, pure pursuit produces finite-time capture and admits no orbit. We will also simulate the classical moving-target pursuit curve for comparison.

- **Model 2: Newtonian gravity model.**

  For the reduced two-body problem,

  $$\ddot{\mathbf{r}} = -\mu \frac{\mathbf{r}}{\rho^3}, \qquad \mu = G(m_1 + m_2), \tag{2}$$

  which, as a first-order system reads

  $$\dot{x} = v_x, \quad \dot{y} = v_y, \quad \dot{v}_x = -\mu \frac{x}{(x^2+y^2)^{3/2}}, \quad \dot{v}_y = -\mu \frac{y}{(x^2+y^2)^{3/2}}.$$

  First, we see that angular momentum is conserved. Let us define the scalar 2D angular momentum $L = xv_y - yv_x = \mathbf{r} \times \dot{\mathbf{r}}$. Then

  $$\frac{dL}{dt} = \dot{\mathbf{r}} \times \dot{\mathbf{r}} + \mathbf{r} \times \ddot{\mathbf{r}} = 0 + 0 = 0,$$

  since $\dot{\mathbf{r}} \times \dot{\mathbf{r}} = 0$ and $\ddot{\mathbf{r}} \parallel \mathbf{r}$.

  Secondly, notice energy is conserved, with the specific energy $E = \frac{1}{2}\|\dot{\mathbf{r}}\|^2 - \mu/\rho$, one finds $dE/dt = 0$, so the system is conservative and does not simply spiral inward. The sign of E classifies the outcome: bounded/elliptic ($E < 0$), parabolic ($E = 0$), and unbounded/hyperbolic escape ($E > 0$).

  Therefore, we show that conserved, generally nonzero angular momentum is precisely what distinguishes orbiting from the collapse seen in pursuit.

- **Model 3: The Generalized Model (Our main research question)**

  With the first two model developed and compared, we then embed both models in a one-parameter family of central forces with potential

  $$V(\rho) = -\alpha\, \rho^{-s}, \qquad \alpha > 0, \tag{3}$$

  so the gravitational case is $s = 1$. Using conservation of L, the radial motion reduces to a one-dimensional energy equation with an effective potential:

  $$E = \tfrac{1}{2}\dot{\rho}^2 + V_{\text{eff}}(\rho), \qquad V_{\text{eff}}(\rho) = V(\rho) + \frac{L^2}{2\rho^2}, \tag{4}$$

  where $L^2/2\rho^2$ is the centrifugal barrier. Whether collision is possible with $L \neq 0$ is a competition between attraction and this barrier as $\rho \to 0$:

  - $s < 2$ (includes gravity): the barrier dominates, $V_{eff} \to +\infty$, and collision is impossible unless $L = 0$.

- $s > 2$: attraction dominates, $V_{eff} \to -\infty$, and collision can occur even with $L \neq 0$ ("fall to the center");
- $s = 2$: the borderline case, whose outcome depends on the sign of $\frac{L^2}{2} - \alpha$.

This yields a clean analytical criterion for collision versus bounded motion versus escape, indexed by the exponent and the angular momentum. We also hope that our work all cleanly build into Bertrand's theorem, which states: among all central-force laws, only the inverse-square law and Hooke's law ($V \propto \rho^2$) produce orbits that close for all bound initial conditions, every other exponent yields complex orbits.

3.2 Mathematical Techniques

We plan to use the following mathematical techniques to analyze and compare Model 1, Model 2 and Model 3:

- Distance function analysis using $\rho(t) = \|\mathbf{r}(t)\|$.
- Comparison between first-order and second-order ODE models.
- Conservation-law analysis for central-force motion, focusing on angular momentum, mechanical energy, and effective potential when appropriate.
- Analyzing different initial conditions to determine their effects on the final behaviour.

3.3 Numerical Methods and Simulations

The framework and models above hand us three concrete, testable predictions: (i) velocity-level pursuit must capture in finite time with no orbit, whereas force-level gravity conserves angular momentum and orbits, (ii) for the generalized family $V(\rho) = -\alpha\rho^{-s}$, the outcome (collision, bounded motion, or escape) is fixed by the interaction between attraction and the centrifugal barrier, with $s = 2$ as the dividing case, and (iii) only special exponents produce closed orbits, while every other exponent precesses (Bertrand). The paper is a computational study that puts these predictions to the test.

The foundation is to implement a system of ODE solvers we can trust. We implement three integrators: forward Euler, classical RK4, and a symplectic velocity-Verlet method, then validate them on the one case with a known exact answer, the Kepler ellipse. Since energy E and angular momentum L are conserved analytically, their numerical drift is a direct, quantitative measure of solver quality: we expect Euler to spiral, RK4 to drift slowly, and the symplectic method to hold a closed orbit over long runs.

With the solvers implemented, the investigation reduces to three experiments:

1. **The dichotomy.** From matched initial conditions, confirm the predicted contrast between pursuit and gravity: linear finite-time capture ($L \equiv 0$) versus a bounded orbit (L conserved and nonzero).
2. **The outcome map.** Sweep the exponent s and the initiale energy and angular momentum (E, L), classify each run as collision, bounded, or escape,and check that the observed boundaries match the centrifugal-barrier criterion.

3. **Bertrand's theorem.** Measure the apsidal (precession) angle as a function of s and verify that orbits close only at the special exponents and precess otherwise.

In conclusion, once we develop the mathematics to tell us what to expect, the paper's computational element will be to measure where the transitions between collision, orbiting, and escape actually occur, and to show that a structure-preserving numerical method reproduces them reliably.

3.4 Theoretical Questions

Throughout our model of gravity, notice that radial acceleration with their initial velocities could change the endpoint of the binary system. Some theoretical questions are:

- 1. How does angular momentum in the Newtonian gravity model which involves angular momentum, prevent a binary system from collapsing, while the pursuit model predict that system will eventually collides? For instance, in pursuit model as time pass it move towards the origin,

$$\dot{\mathbf{r}} = -v\,\hat{\mathbf{r}} \tag{5}$$

  For second model, if we consider that there is a velocity that is pointing towards sideway respect to direction of gravitation pull,

$$\ddot{\mathbf{r}} = -\mu\,\frac{\mathbf{r}}{\rho^3} \implies L = xv_y - yv_x \neq 0 \tag{6}$$

  which we can conclude that these two model's outcomes are not directly connected. So we are going to investigate about what is the mathematical difference between these two model.

- 2. For model 3 and 1 We discovered that the both of model explains the causation of the collision but differently. In model 1, We already mentioned that the velocity is force to be radial thus the angular momentum is zero. But in model 3, collision only occurs when

$$L \neq 0, V(\rho) < \frac{L^2}{2\rho^2} \tag{7}$$

  Which we can conclude our new theoretical question, what mainly cause the collision in the binary system? Due to central attraction strength or angular momentum?

4 Project Resources

Our software tools will include:

- Python with NumPy (numerics), SciPy `solve_ivp`, and Matplotlib.
- Git/GitHub for collaboration.
- REBOUND for high-accuracy symplectic integrators.

References

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5 Timeline and Milestones

- **Week 1:** We will further refine the model and investigate potential parameters and models. Specifically, firstly, we will examine whether the pursuit model is suitable as a baseline, and whether it has any other potential roles beyond serving as a baseline; regarding the central force model, varying the value of the force exponent p will enable us to study how changes in the relationship between gravitational force and distance affect collisions, bounded motion and escape; furthermore, if there are other forms of 'energy loss', we will investigate whether the original system would result in a spiral sink. Throughout these processes, we will gather further references to assist us in identifying factors and updating the model.

  After identifying the parameters, we will investigate the comparison between collision, orbit and escape. Furthermore, if we use numerical methods, such as Euler's method, we will also need to investigate their accuracy.

- **Week 2:** Once the parameters have been determined and preliminary feasibility tests have been carried out, we need to derive the final model. Once the pursuit model has been established, we can analyze why this model leads to collision. If we want to do the numerical simulation in future work, we need to write it in the first-order system. In addition, we will need to determine various initial conditions (ICs) that are expected to result in collision, bounded orbit, and escape, in order for the comparison.

- **Week 3:** Once we have obtained the formal model, we will run it in a programming environment (such as Python) to generate preliminary images. After that, we will test different ICs to determine which scenarios lead to collision, bounded orbital motion, or escape, and compare the differences between the models. In addition, we will commence numerical simulations: by utilising and comparing the stability and time intervals of various numerical methods, we will determine the optimal method.

- **Week 4 and 5:** After numerical simulations, we will experiment with the models repeatedly and more systematically. For instance,the for gravity model, we will change the initial velocities to observe the possible changes of qualitative behaviours, like collision, orbit or escape. Then, we will compare these results to the pure pursuit model, as it may result in collision.

  Also, we may use the generated data visualizations, such as $\|r(t)\|$, trajectory graphs and angular momentum to help us analyze. For gravitational model, since our model is based on the assumption of Newtonian two-body system, we will check if the modelling behaviour satisfies the assumption.

  In the end, we will choose informational data result and figures to compare the differences between the models that pursuit model is based on velocity-directed motion whilst gravity model is based on acceleration-directed motion. Finally, we will finish our final report to draw the conclusion of discussing in which conditions it will result in collision, orbit or escape.